

MSE 311 Thermodynamics of materials-Homework 5

Name:

Student ID:

Please hand in this homework before class on October 13th.

*Suggest read the reference book **Thermodynamics: an Engineering Approach** after class.*

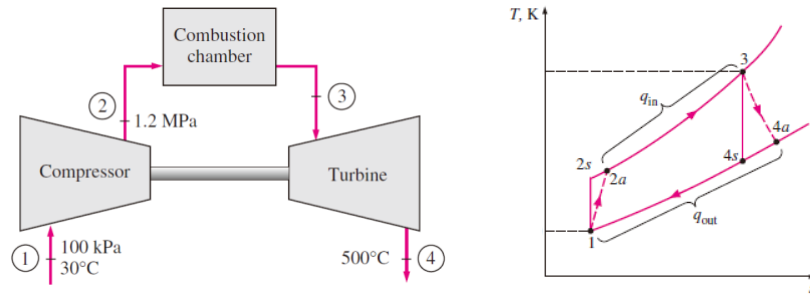
1. Derive the following efficiency formula of Stirling cycle or Ericsson cycle.

$$\eta = 1 - \frac{T_L}{T_H}$$

2. Search the maximum allowable temperature for the current alloy, and give the alloy name and the related literature.

3. A gas-turbine power plant operates on the simple Brayton cycle between the pressure limits of 100 and 1200 kPa. The working fluid is air, which enters the compressor at 30°C at a rate of 150 m³/min and leaves the turbine at 500°C. Using variable specific heats for air and assuming a compressor isentropic efficiency ($\eta_c = (h_{2s} - h_1) / (h_{2a} - h_1)$) of 82 percent and a turbine isentropic efficiency ($\eta_t = (h_3 - h_{4a}) / (h_3 - h_{4s})$) of 88 percent, determine (a) the net power output, (b) the back work ratio, and (c) the thermal

efficiency.



4. Consider a steam power plant that operates on a simple ideal Rankine cycle and has a net power output of 45 MW. Steam enters the turbine at 7 MPa and 500°C and is cooled in the condenser at a pressure of 10 kPa by running cooling water from a lake through the tubes of the condenser at a rate of 2000 kg/s. Show the cycle on a T-s diagram with respect to saturation lines, and determine (a) the thermal efficiency of the cycle, (b) the mass flow rate of the steam, and (c) the temperature rise of the cooling water.

5. The van der Waals gas constant of CO₂ gas is $a = 0.3638 \text{ Pa} \cdot \text{m}^6 \cdot \text{mol}^{-2}$, $b = 4.3 \cdot 10^{-5} \text{ m}^3 \cdot \text{mol}^{-1}$, and the boiling point of CO₂ is 194.7 K. When the gas expands from $50 \cdot p^0$ to $1 \cdot p^0$ through a throttle hole, its temperature drops from 298 K to 234 K. (2019)

$$\mu_{J-T} = \frac{1}{C_{p,m}} \left(\frac{2a}{RT} - b \right)$$

(1) What is the conversion temperature of the above CO₂ gas (The conversion temperature is the

temperature when $\mu_{J-T}=0$).

(2) When the above CO₂ gas is throttled and expanded further to make its temperature down to the boiling point (the final pressure is still $1 \cdot p_0$), what is the initial pressure?

6 Finishing the derivation of $\eta_{\max}$ for a thermoelectric generator, and thinking about whether the Joule heat term in the Q_c could be less than $\frac{1}{2} I^2 R$, if possible how to achieve it, if not why?